RC PLANE TASK 2:Paper Airplane Challenge: Designing for Maximum Flight Time

1. Assumptions

- **Paper:** A4, 297 × 210 mm, 80 g/m² (check the packaging). Mass ≈ 0.297 × 0.210 × 80 ≈ **5.0 g**.
- **Air:** still indoor air, density 1.225 kg/m³.
- **Flight:** time is measured from the moment you release the plane until it first touches the floor.
- **Flow regime:** at a glide speed of about 2 m/s and a wing chord of about 0.2 m, the Reynolds number is $Re = V \cdot c/\nu \approx 2 \times 0.2 / 1.46\times10^{-5} \approx$ **27,000**. This is very low, so the paper behaves like a thin flat plate, not like a thick airfoil.
- **Coefficients:** the lift and drag values used in calculations are assumed, not measured.

2. Materials

- At least 6 sheets of A4 from the same pack (so they are identical), a ruler, a pencil and a kitchen scale (0.1 g if possible).
- A phone for slow-motion video and timing, plus a tape measure.
- Chalk or masking tape for marking a launch line **on the floor** (not on the plane).

3. Part A: How to Build the Planes

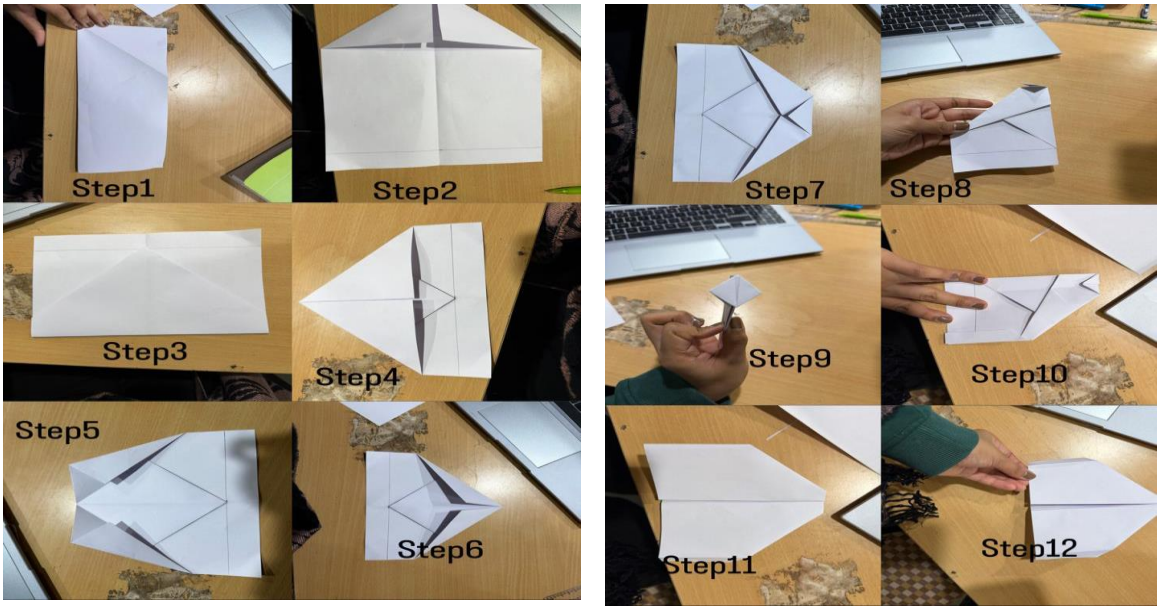
The two designs share the first folds on purpose. This makes the later comparison a fair test, because only the differences I list can explain different results.

Design A: Wide-Wing Glider (the main design)

Hold the paper **portrait** (210 mm wide, 297 mm tall). Crease every fold firmly with a fingernail or ruler, because sloppy folds make the plane fly differently every time.

1. **Centre crease.** Fold in half lengthwise, then unfold. *Why: this line is the plane's backbone, and every later fold is measured from it. A plane that is not symmetrical will veer to one side.*
2. **Fold the top corners in.** Fold both top corners so the top edge lies along the centre crease. You get a pointed triangle at the top. The line across the paper where the two slanted folds meet the outer edges is the **shoulder line** (about 105 mm from the top). *Why: this makes the nose and gives the plane its basic shape.*
3. **Fold the tip down.** Fold the pointed tip down so it lands on the shoulder line. You now have a flat-fronted nose. *Why: it stacks extra layers of paper at the front, which adds weight in the right place.*
4. **Fold the flat front edge back.** Fold the new front edge back again so it also lies on the shoulder line. *Why: more layers at the nose means more weight forward. That moves the centre of gravity forward, which makes the plane stable (Section 5.3). It also stiffens the leading edge so the wing does not flutter.*

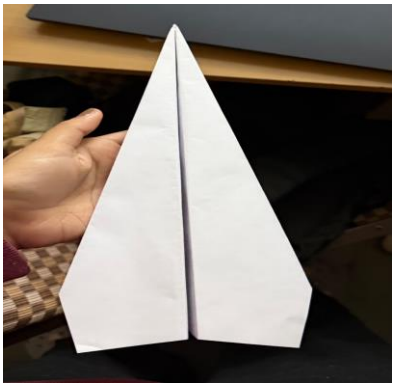
5. **Fold in half.** Fold along the centre crease so all the folds you have made stay on the **outside**. *Why: this makes the central keel.*
6. **Fold the wings down.** Fold each wing down along a line parallel to the bottom edge, about **15 mm** above it. The 15 mm strip is the keel, and each wing is now about 90 mm wide. *Why: wide wings mean large wing area, which is the main reason this design stays up longer than a dart. The keel adds stiffness and helps keep the plane pointing straight.*
7. **Set the shape.** Open the wings so they form a very shallow V when viewed from the front, about 10°. Wings should be flat, not twisted. *Why: this is dihedral, and it makes the plane roll itself level after a gust or a bad launch.*
8. **Add trim.** Bend the rear edge of each wing up by 2 to 3 mm over its whole width. *Why: this small lip lifts the tail, so it prevents the nose from drooping and gives a gentle, flat glide. You will adjust it during testing.*



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Design B: Classic Dart (for the bonus comparison)

1. Fold in half lengthwise, then unfold.
2. Fold both top corners to the centre crease.
3. Fold the new slanted edges to the centre crease **again**, making a narrow pointed nose.
4. Fold in half along the centre crease, keeping the folds on the outside.
5. Fold each wing down about **25 to 30 mm** from the bottom edge, and open the wings flat.



Why it is the right comparison: it is the standard design that most people mean by a paper plane, built for speed and distance. It has smaller wings and a narrower, heavier-looking nose.

4. Why These Design Choices

Choice	What I did	Reason
Wing shape	Wide, low-aspect-ratio, delta-like wing from the dart fold	Easy to build with one sheet and no glue. It is stable and flies slowly. A long thin wing would be more efficient but very hard to make from one folded sheet.
Wing area	As large as one A4 allows (about 0.039 m^2, assumed, measure yours)	Lower wing loading gives lower flight speed and lower sink rate (Section 5.4).
Weight	Nose folded into several layers, no added objects	Paper only, as required. The weight is where it is needed (forward).
Configuration	Central keel, shallow V, small trailing-edge lip	Stability, straight flight and pitch trim without tape or glue.
Control surfaces	Hand-bent trailing edges	They let you adjust the plane's behaviour in seconds.
Launch method	Gentle upward lob from head height or higher	Height is stored energy. See Section 5.2.

5. Part B: Aerodynamics

5.1 How does the plane make lift?

Lift is the upward force that holds the plane in the air. It comes from the wing pushing air downward and from the pressure difference between the upper and lower surfaces.

- **Newton's explanation:** the tilted wing deflects the air it meets downward. The air is pushed down, so the wing is pushed up.
- **Pressure explanation:** the airflow over the top is curved and sped up, so the pressure there falls. Pressure under the wing is higher. The difference over the wing area adds up to lift.

These are two views of the same thing. A paper plane has a thin, nearly flat wing, so most of its lift comes from **angle of attack**: the angle between the wing and the oncoming air. More angle gives more lift, up to the **stall** (at about 12 to 15° for a flat plate, assumed), where the air separates from the upper surface and lift collapses.

5.2 Four forces: launch and glide

Force	Direction	During launch	During glide
Weight ($W = mg \approx 0.005 \times 9.81 \approx 0.049$ N)	Down	Constant	Constant
Lift	Perpendicular to the flight path	Builds with speed	Balances most of the weight
Drag	Opposite to the motion	High at high launch speed	Slows the plane
Thrust	Forward	Your arm provides it for about 0.2 s	**None.** There is no engine.

How it keeps flying without thrust: at release, your throw gives the plane energy. A throw of 6 m/s gives kinetic energy $\frac{1}{2} m v^2 = 0.5 \times 0.005 \times 6^2 \approx 0.09$ J, which could lift the plane about $v^2/2g \approx 1.8$ m if none were lost to drag. After release, the plane slopes downward and a part of its own weight ($W \sin \gamma$, where γ is the glide angle) pulls it forward along the flight path. So **gravity does the job of thrust**, and the plane pays for its flight by slowly losing height.

In a steady glide: $L = W \cos \gamma$ and $D = W \sin \gamma$, so $\tan \gamma = D/L = 1/(L/D)$.

Important consequence: flight time ≈ starting height ÷ sink rate. To stay up longer you either start higher or sink more slowly. Design A reduces the sink rate, and a high upward launch increases the starting height.

5.3 Wing area, wing loading, centre of gravity, stability

- **Wing area (S):** the planform area of the wings. Measure it by tracing the outline on squared paper and counting squares.
- **Wing loading (W/S):** weight divided by wing area. Using the assumed areas, the glider is about $0.049 / 0.039 \approx$ **1.3 N/m²** and the dart about $0.049 / 0.020 \approx$ **2.5 N/m²** (both areas assumed, so measure yours). Lower wing loading means the plane can fly slower, because the speed needed to hold it up is $V = \sqrt{(2W / \rho S C_L)}$. With $C_L \approx 0.7$ (assumed): glider ≈ 1.7 m/s, dart ≈ 2.4 m/s.
- **Centre of gravity (CG):** the point where the plane balances. **[YOU]** Find it by resting the plane on two fingertips under the wings and sliding them until it balances level. Mark the spot and

measure its distance from the nose. As a starting target, aim for the balance point about **one-third of the way back from the nose**, then tune from test glides.

- **Stability:** a plane is *pitch-stable* if, when a gust raises the nose, it automatically tends to lower it again. This works when the CG is **ahead** of the wing's centre of lift (the aerodynamic centre). Too far forward and the plane dives. Too far back and it is unstable and stalls or tumbles. *Dihedral* (the shallow V) gives roll stability, and the *keel* resists sideways yawing.

5.4 Why this design should outlast a dart: minimum sink vs. maximum glide

A dart is built to fly **fast and far**: small wings, high wing loading, high speed. A fast plane covers a lot of ground but loses height quickly, so it is in the air for only a short time. Design A is built to fly **slow and floaty**.

Two different goals need two different speeds:

- **Maximum glide distance** (flying farthest from a given height): fly at the speed where **L/D is highest**. This is the flattest glide angle.
- **Minimum sink rate** (staying up longest): fly at the speed where the **vertical descent speed is lowest**. This is **slower** than the best-glide speed.

The sink rate is

sink = V / (L/D) = √(2W / ρS) × C_D / C_L^1.5

So to stay up longest you want low wing loading (small W/S) **and** a high C_L^1.5 / C_D, meaning you fly slowly at a fairly high angle of attack, just below stall. A plane can have an excellent glide ratio and still have a poor sink rate if it flies fast.

Illustrative calculation (assumed values): if both planes had L/D = 4:

- Glider: sink ≈ 1.7 / 4 ≈ 0.43 m/s, so from 2 m height ≈ **4.7 s**.
- Dart: sink ≈ 2.4 / 4 ≈ 0.60 m/s, so from 2 m height ≈ **3.3 s**.

This only shows the *direction* of the effect. Real values depend on real L/D, and the dart may actually have a better L/D than the glider. Your measurements decide.

6. Part C: Experiment Protocol

Goal: make every flight as identical as possible, so differences in time come from the plane, not from the person or the room.

Controlled (kept the same): one thrower, same launch spot marked on the floor, same release height (measure it, about 2 m), same launch angle (about 30° to 45° upward), same gentle launch strength, same room with doors and fans off, same batch of paper, same stopwatch method.

Measured (dependent variable): flight time, from release to first floor contact.

How to time reliably: film each flight with the phone and read the time from the video frame by frame (or use two stopwatches and average them). A human stopwatch alone has a reaction error of about 0.2 to 0.3 s, which is large for flights of a few seconds.

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7. Trial Table and Results

Trial	Release height (cm)	Flight time (s)	flight distance(cm)
1	150cm	5.2s	696cm
2	152cm	5s	688cm
3	151cm	4.8	652cm
4	152cm	4.3s	644cm
5	152cm	5.8s	712cm
Mean	151.4 cm	5.02 s	678.4 cm

Calculations:

- **Mean** = (sum of all times) ÷ number of flights.
- **Spread (standard deviation)** = how far the individual times scatter around the mean. In a spreadsheet: STDEV.S.
- **Standard error** = standard deviation ÷ √(number of flights).

Reading the result: if two designs (or two versions) differ by less than about two standard errors, you cannot say one is truly better. Say so honestly, and do more trials.

Worked results for your five trials (Design A)

Quantity	Release height (cm)	Flight time (s)	Flight distance (cm)
Mean	151.4	5.02	678.4
Standard deviation	0.9	0.55	29.2
Standard error	0.4	0.25	13.1
Shortest / longest	150 / 152	4.3 / 5.8	644 / 712

Worked example (flight time):

1. Add the times: 5.2 + 5.0 + 4.8 + 4.3 + 5.8 = 25.1 s.
2. Mean = 25.1 ÷ 5 = **5.02 s**.
3. Difference of each flight from the mean: +0.18, −0.02, −0.22, −0.72, +0.78.
4. Square each, add them (1.208), divide by n − 1 = 4 (0.302), then take the square root: standard deviation = **0.55 s**.
5. Standard error = 0.55 ÷ √5 = **0.25 s**.

The same steps give the height and distance columns.

Result: mean flight time **5.0 ± 0.25 s** (mean ± standard error). The flights vary by about 11% (0.55 ÷ 5.02).

Simple extra calculations (rough, because the throw adds some height and speed):

- Average sink rate ≈ release height ÷ flight time = 1.514 m ÷ 5.02 s ≈ **0.30 m/s**.
- Average forward speed ≈ distance ÷ time = 6.784 m ÷ 5.02 s ≈ **1.35 m/s**.
- Distance ÷ release height = 678.4 ÷ 151.4 ≈ **4.5**, a rough glide ratio.

What it tells you:

- The release height was very consistent (150 to 152 cm), so the launch was well controlled.
- The flight times range from 4.3 s to 5.8 s, which is more than the standard error. Five flights is a small sample, so do 10 if you can.
- The earlier illustration assumed a 2 m release and a sink rate of 0.43 m/s (4.7 s). You measured a lower sink rate of about 0.30 m/s from 1.5 m. This suggests the glider sinks more slowly than assumed, or that the lofted launch added height. State this honestly and do not claim more than the data show.

8. Two Modifications and How to Test Them

Rule: **change one thing at a time**, so you know what caused any change.

Modification1: cutting the paper by 2 cm from bottom.

When I cut 2cm off from the bottom, the flight time showed a slight decrease. Cutting off paper from the bottom helped balance the center of gravity but it also decreased the wing area thus resulting in lesser lift force. The reason behind trying this modification is that I noticed that during most of the trial flights, the paper plane dived steeply before coming to a halt.

Modification2:change the nose weight. Add one extra fold at the nose. Reason: this moves the CG, and the best CG for a long flight is a balance between stability and flight attitude.

Adding an extra fold causes the center of gravity to shift towards the nose which hence making the flight time significantly lesser and making the plane crash much faster.

9. Bonus: Design A vs. Design B (Dart)

1. Build the dart (Section 3) from the same batch of paper.
2. Use the **same protocol** for both. Alternate A, B, A, B, … so that room conditions and your arm tiredness affect both equally. Fly at least 5 flights each, 10 if possible.
3. Launch rule: same height and angle for both. The dart needs a firmer throw to fly properly, so state this clearly in the report. If you want to be thorough, also test the dart at its own best launch speed.
4. Present the means, standard deviations and a short conclusion in a small table.

Design	Mean time (s)	Number of trials
A: Wide-wing glider	5.02	5

Design	Mean time (s)	Number of trials
B: Classic dart	3.4	5

10. Troubleshooting and Trim Guide

What you see	Likely cause	Try this
Dives steeply, nose-first	Nose too heavy, or trailing edge not lifted	Bend the trailing edges up a little more, or take out a nose fold.
Climbs, stalls, then drops	Nose too light, or trailing edge lifted too much	Bend the trailing edges down a little, or add a nose fold.
Veers to one side	Wings not symmetrical, or a twisted wing	Check the folds and unbend any twist. A tiny trailing-edge bend on one wing can correct a drift (if it veers right, bend the left rear edge slightly up; if it gets worse, do the opposite).
Rolls over	Too little dihedral	Open the V a little more.
Flies fast and flat, short flight	Too hard a launch	Throw gently and upward.